TECHNICAL DOCUMENTATION

SPHEREx Motion Explorer

Architecture, data flow, and the science core: bit-correct SPHEREx flag masking, per-pixel spectral-WCS wavelengths, parallel aperture spectrophotometry, IRSA AllWISE reference imagery, and a zero-build frontend.

SPHEREx Motion Explorer team - v0.5.0 -

November 2026

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1. System Architecture

A three-tier, zero-database design: the browser renders everything; a same-origin FastAPI process brokers all archive traffic (adding retries, allowlists, and bounded caches); NASA/IPAC IRSA remains the only source of science data. No invented data exists anywhere in the pipeline.

Tier	Components	Responsibility
Browser	Vanilla HTML/CSS/JS, Aladin Lite v3 (vendored, 2.4 MB, WebGL2)	Workbench UI, epoch timeline, WCS canvas overlays, zoom/pan, SVG spectrum plot
API server	FastAPI + uvicorn + httpx (Python 3.12)	Six endpoints brokering IRSA SIA/IBE/TAP; retries with backoff; SSRF allowlists; bounded LRU caches
Science core	Astropy 8.0.2 WCS, reproject 0.21, numpy, Pillow	Flag-bit masking, TAN reprojection, aperture photometry, spectral-WCS interpolation, mini-FITS generation
Archive	NASA/IPAC IRSA - SIA v2, IBE (search + cutouts), TAP	SPHEREx QR2/QR3 exposure metadata and FITS cutouts; AllWISE coadds; Gaia DR3

Browser → FastAPI (same origin) → IRSA → FITS bytes → Astropy masking + reprojection → PNG/JSON → browser canvas. Every response carries provenance headers.

2. Backend Modules

Module	Lines	Responsibility
backend/archive.py	~140	SIA v2 CSV parsing (Caltech/IPAC format), per-record wavelength ranges, great-circle off-axis angles, UTC-to-MJD TIME filters, 3-attempt retry
backend/flags.py	~80	Official 32-bit flag semantics from the Explanatory Supplement; strict vs display mask levels; per-bit provenance summaries
backend/science.py	~330	Cutout fetch (size-padded, cached), fatal-bit masking, VARIANCE/ZODI extraction, local-wavelength interpolation, reprojection to a shared TAN grid, stretch/difference/sequence rendering, wavelength-match verdicts
backend/spectrum.py	~180	Parallel aperture spectrophotometry across up to 60 files; adaptive strict-to-display mask fallback for saturated cores; even wavelength sampling preferring target-centred frames
backend/catalog.py	~110	IRSA TAP Gaia DR3 queries (table gaia_dr3_source), proper-motion propagation to the observation epoch, VOTABLE error surfacing
backend/atlas.py	~150	AllWISE coadd reference imagery: IBE search (POS=ra,dec), tile selection, gzip cutout, asinh display FITS with preserved WCS

Module	Lines	Responsibility
backend/app.py	~150	FastAPI wiring: /api/health, /observations, /render, /sequence, /spectrum, /catalog, /atlas_image, /net_check; coordinate validation; static frontend mount

3. The Science Core

3.1 Bit-correct flag masking

SPHEREx L2b FLAGS extensions are 32-bit status bitmaps. The naive flags != 0 rule masks bit 21 (SOURCE - pixels containing real star flux, set on ~84 percent of pixels in typical fields) and produced 0.15 percent valid coverage: blank images. Our mask follows the official bit table (SSDC-DP-017 v2.0, pages 31-32):

Class	Bits	Treatment
Fatal (strict)	TRANSIENT 0, OVERFLOW 1, SUR_ERROR 2, NONFUNC 6, DICHROIC 7, MISSING_DATA 9, HOT 10, COLD 11, PHANMISS 14, NONLINEAR 15, PERSIST 17, OUTLIER 19, CROSSTALK 20, GHOST 22-24, STREAK 25, BLOOM 26, SNOWBALL 27, HALO 28, SATELLITE_HALO 29	Masked for display and photometry
Display-keep	saturated-core family (0/1/2/15/17/19/20/22-29)	Kept visible in visual modes so bright stars are not holes; strict level still masks them for photometry
Always visible	SOURCE 21, FULLSAMPLE 12	Flux-bearing and informational - never masked

3.2 Per-pixel local wavelength

SPHEREx uses a linear variable filter: the wavelength sampled at a sky position depends on where it lands on the detector, so two exposures of the same detector band can sample different wavelengths. The tool maps the target through the exposure's celestial WCS to a cutout pixel, converts via the alternate LINEAR 'A' WCS (0-based active-pixel coordinates, CRPIX*A) to the original detector frame, and bilinearly interpolates the 9x9 WCS-WAVE lookup grid. Verified result: lambda = 1.728 vs 2.224 um for two D3 frames (a 0.50 um mismatch the file-level metadata cannot reveal), and 0.776 vs 1.005 um across two D1 epochs.

3.3 Exploratory spectrophotometry

Each AllWISE-style pass extracts r = 2 pixel aperture sums with a 3-6 pixel annulus background from up to 60 evenly wavelength-sampled files (6 concurrent downloads, 3-attempt retries). Errors are the quadrature sum of

the VARIANCE extension over the aperture. Bright sources saturate SPHEREx cores; when the strict mask empties the aperture, extraction retries with the display mask and flags the point saturated_core=true (a labelled lower limit). Verified run: 40/40 points, 0.795-4.836 um, SNR 175-406, all six detectors represented.

4. Frontend Engineering

• Zero build step: three static files (index.html, app.js, style.css) served by FastAPI - a volunteer can read every line.

• Aladin Lite v3 vendored locally (aladin-lite@3.8.2, ES module with inlined WASM) and published as window.A via a module shim - no CDN dependency, works on filtered networks.

• Gaia overlay: epoch-propagated star positions projected through the display grid WCS into per-frame canvases (split-mode aware), with proper-motion arrows exaggerated 50x and a translucent caption strip.

• Frame zoom/pan: CSS transforms on wrapper divs containing both the science image and its overlay canvas, so circles, rings, and motion arrows stay locked to their stars at any zoom; wheel, buttons, and pointer-drag pan.

• Two-layer reference atlas: a sharp field cutout over a ~1.25-deg AllWISE context image, both generated server-side as WCS-preserved mini-FITS and displayed with Aladin displayFITS.

• URL deep links (ra/dec/fov/release/survey/band/a/b/mode/sequence/view) reproduce any view; evidence manifests export observation IDs, spectra, caveats, and the dataset DOI.

5. API Reference

Endpoint	Purpose	Key parameters
GET /api/health	Service + scientific-engine status	-
GET /api/observations	IRSA SIA v2 exposure discovery	ra, dec, fov, release, survey, start_date, end_date
GET /api/render	Registered PNG frames + provenance	ra, dec, fov, a, b, mode=a|b|mean|difference
POST /api/sequence	2-8 frame chronological playback	JSON body: ra, dec, fov, tokens[]
GET /api/spectrum	On-the-fly spectrophotometry	ra, dec, fov, release, survey, max_files (4-60)
GET /api/catalog	Gaia DR3 overlay, epoch-propagated	ra, dec, fov, mag_limit, epoch
GET /api/atlas_image	AllWISE reference mini-FITS	ra, dec, fov, ref=wise, tier=detail|context
GET /api/net_check	Tile-host reachability probe	host (allowlisted CDS hosts)

6. Reliability, Security, Testing

• **Retries with backoff:** SIA x3, cutouts x3, TAP x2, IBE x3 - the IRSA route intermittently stalls on some networks; every layer degrades honestly instead of fabricating content.

• **SSRF hardening:** image sources are base64url tokens validated against an allowlist (https, irsa.ipac.caltech.edu host, /ibe/data/spherex/ path, .fits extension, no query/credentials). Net-check hosts are additionally allowlisted.

• **Bounded resources:** cutouts capped at 28 MB, reference tiles at 60 MB, sequences at 8 frames, spectrum sweeps at 60 files, LRU caches on cutouts (48), frames (24), and atlas products (16).

• **Testing:** 29 offline unit tests cover flag semantics (bit 21 never masked), palette-free rendering, aperture photometry on synthetic frames, spectral-WCS interpolation, file sampling, API validation, and the HTML/JS id contract.

7. Deployment

The Dockerfile (python:3.12-slim) runs uvicorn on $PORT as an unprivileged user - deployable to any container host. For zero-cost demonstrations, start_public.bat launches the server plus a Cloudflare quick tunnel and prints a public HTTPS URL. Requirements: Python 3.11/3.12 or Docker; outbound access to irsa.ipac.caltech.edu (CDS HiPS optional for the optical surveys).

```
docker build -t spherex-motion-explorer . → docker run -p 7860:7860 spherex-motion-explorer
```

8. Known Limitations

• WCS-WAVE wavelengths are visualization-grade (the supplement recommends the separate spectral-WCS files for science analysis).

• No PSF matching, aperture correction, or absolute calibration; difference modes remain visual triage aids.

• SIA discovery caps at 600 records per query (pagination on the roadmap).

• Reference-atlas context is bounded by the AllWISE coadd tile (~1.25 deg); full-sky zoom-out requires CDS HiPS reachability.